



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science  
Faculty of Computing

## Project Task 5

( *IP ADDRESSING SCHEME* )

|                     |                                                                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programme</b>    | Bachelor of Computer Science ( <i>Data Engineering</i> )                                                                                         |
| <b>Subject Code</b> | SECR1213                                                                                                                                         |
| <b>Subject Name</b> | Network Communications                                                                                                                           |
| <b>Session-Sem</b>  | 2024/2025-1                                                                                                                                      |
| <b>Prepared by</b>  | 1) ANIS SAFIYYA BIN JANAI<br>2) NURUL ASYIKIN BINTI KHAIRUL ANUAR<br>3) DAMIYA AINA BINTI BASIR ABD SHAMMAD<br>4) AIN NURNABILA BINTI MOHD AZHAR |
| <b>Section</b>      | 02                                                                                                                                               |
| <b>Group</b>        | CircuitCrew (CC)                                                                                                                                 |
| <b>Lecturer</b>     | DR. ISMAIL FAUZI BIN ISNIN                                                                                                                       |
| <b>Date</b>         | 28 / 12 / 2024                                                                                                                                   |

## Table of Content

|                                          |          |
|------------------------------------------|----------|
| <b>Get IP Address from Lecturer.....</b> | <b>3</b> |
| <b>Divide the Subnet Equally.....</b>    | <b>5</b> |
| <b>Cable type and length.....</b>        | <b>7</b> |
| <b>Minutes meeting #1.....</b>           | <b>8</b> |
| <b>Minutes Meeting #2.....</b>           | <b>9</b> |

## 1.0 Get IP Address from Lecturer

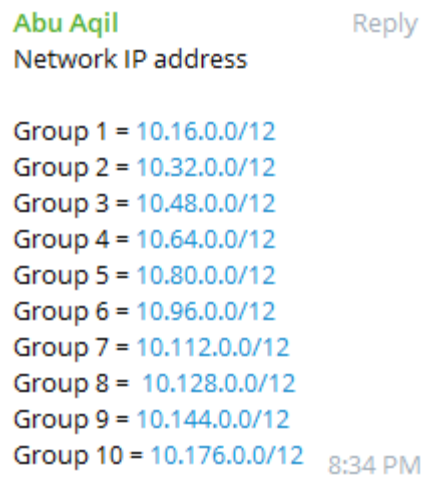


Figure 1.0.1

Our group got the network address of 10.96.0.0/12. To ensure that our range of address usage does not conflict with other groups' subnets, we make the assumption that the subnet mask for our network is /12 or 255.240.0.0.

Network Address Details:

|                       |                                     |
|-----------------------|-------------------------------------|
| IP Address (decimal)  | 10.96.0.0                           |
| IP Address (binary)   | 00001010 01100000 00000000 00000000 |
| Subnet Mask (decimal) | 255.240.0.0                         |
| Subnet Mask (binary)  | 11111111 11110000 00000000 00000000 |

With the subnet mask, we divided the subnet mask into the network portion (12 bits) and host portion (20 bits). Then, we determined the network address using the **AND** operation of the IP address with the subnet mask. The broadcast address is calculated by setting all host bits to 1.

|                         |                            |
|-------------------------|----------------------------|
| Network Address         | 10.96.0.0                  |
| Broadcast Address       | 10.111.255.255             |
| Range of usable address | 10.96.0.1 - 10.111.255.254 |

## 2.0 Divide it in the best possible way for your network.

Using the classful addressing techniques, we choose to divide the network into several subnets based on the task 4 work area and an extra subnet for our server room. As a result, our building's network requires a total of 9 subnets. This is managed by borrowing an extra 4 bits ( $2^2 = 4$ ) from the host portion in order to form the subnet.

| Subnet | Network Portion   | Subnet Bits | Host Portion           |                   | IP Address     |
|--------|-------------------|-------------|------------------------|-------------------|----------------|
| 0      | 0000 1010<br>0110 | 0000        | 0000 0000 0000<br>0000 | Network Address   | 10.96.0.0      |
|        |                   |             | 1111 1111 1111<br>1111 | Broadcast Address | 10.96.255.255  |
| 1      | 0000 1010<br>0110 | 0001        | 0000 0000 0000<br>0000 | Network Address   | 10.97.0.0      |
|        |                   |             | 1111 1111 1111<br>1111 | Broadcast Address | 10.97.255.255  |
| 2      | 0000 1010<br>0110 | 0010        | 0000 0000 0000<br>0000 | Network Address   | 10.98.0.0      |
|        |                   |             | 1111 1111 1111<br>1111 | Broadcast Address | 10.98.255.255  |
| 3      | 0000 1010<br>0110 | 0011        | 0000 0000 0000<br>0000 | Network Address   | 10.99.0.0      |
|        |                   |             | 1111 1111 1111<br>1111 | Broadcast Address | 10.99.255.255  |
| 4      | 0000 1010<br>0110 | 0100        | 0000 0000 0000<br>0000 | Network Address   | 10.100.0.0     |
|        |                   |             | 1111 1111 1111<br>1111 | Broadcast Address | 10.100.255.255 |
| 5      | 0000 1010<br>0110 | 0101        | 0000 0000 0000<br>0000 | Network Address   | 10.101.0.0     |
|        |                   |             | 1111 1111 1111<br>1111 | Broadcast Address | 10.101.255.255 |
| 6      | 0000 1010<br>0110 | 0110        | 0000 0000 0000<br>0000 | Network Address   | 10.102.0.0     |
|        |                   |             | 1111 1111 1111<br>1111 | Broadcast Address | 10.102.255.255 |
| 7      | 0000 1010<br>0110 | 0111        | 0000 0000 0000<br>0000 | Network Address   | 10.103.0.0     |

|   |                   |      |                        |                      |                |
|---|-------------------|------|------------------------|----------------------|----------------|
|   |                   |      | 1111 1111 1111<br>1111 | Broadcast<br>Address | 10.103.255.255 |
| 8 | 0000 1010<br>0110 | 1000 | 0000 0000 0000<br>0000 | Network Address      | 10.104.0.0     |
|   |                   |      | 1111 1111 1111<br>1111 | Broadcast<br>Address | 10.104.255.255 |

| Subnet       | Work Area                     | Network<br>Address | Broadcast<br>Address | Range of usable<br>address     | Custom subnet<br>mask |
|--------------|-------------------------------|--------------------|----------------------|--------------------------------|-----------------------|
| 0            | Server Room                   | 10.96.0.0          | 10.96.255.255        | 10.96.0.1 -<br>10.96.255.254   | /16<br>255.255.0.0    |
| Ground Floor |                               |                    |                      |                                |                       |
| 1            | General-purpo<br>se Lab       | 10.97.0.0          | 10.97.255.255        | 10.97.0.1 -<br>10.97.255.254   | /16<br>255.255.0.0    |
| 2            | Video<br>Conferencing<br>Room | 10.98.0.0          | 10.98.255.255        | 10.98.0.1 -<br>10.98.255.254   | /16<br>255.255.0.0    |
| 3            | Student<br>Lounge             | 10.99.0.0          | 10.99.255.255        | 10.99.0.1 -<br>10.99.255.254   | /16<br>255.255.0.0    |
| 4            | General-purpo<br>se Lab I     | 10.100.0.0         | 10.100.255.255       | 10.100.0.1 -<br>10.100.255.254 | /16<br>255.255.0.0    |
| 5            | General-purpo<br>se Lab II    | 10.101.0.0         | 10.101.255.255       | 10.101.0.1 -<br>10.101.255.254 | /16<br>255.255.0.0    |
| First Floor  |                               |                    |                      |                                |                       |
| 6            | CISCO Lab                     | 10.102.0.0         | 10.102.255.255       | 10.102.0.1 -<br>10.102.255.254 | /16<br>255.255.0.0    |
| 7            | Embedded<br>Lab               | 10.103.0.0         | 10.103.255.255       | 10.103.0.1 -<br>10.103.255.254 | /16<br>255.255.0.0    |
| 8            | Hybrid<br>Classroom           | 10.104.0.0         | 10.104.255.255       | 10.104.0.1 -<br>10.104.255.254 | /16<br>255.255.0.0    |

## Meeting Minutes #1

|                     |                                       |                                                         |                                |
|---------------------|---------------------------------------|---------------------------------------------------------|--------------------------------|
| <b>Date/Time</b>    |                                       | 19 Dis 2024 / 5.00 pm                                   |                                |
| <b>Location</b>     |                                       | Foyer Dewan Resak                                       |                                |
| <b>Agenda</b>       |                                       | Task Distribution for Task 5                            |                                |
| <b>Meeting Host</b> |                                       | Nurul Asyikin binti Khairul Anuar                       |                                |
| <b>Attendance</b>   |                                       |                                                         |                                |
| <b>Name</b>         |                                       | <b>Time</b>                                             | <b>Reason For Absence</b>      |
| Anis Safiyya        |                                       | 5.00 pm                                                 | -                              |
| Nurul Asyikin       |                                       | 5.00 pm                                                 | -                              |
| Damiya Aina         |                                       | 5.00 pm                                                 | -                              |
| Ain Nurnabila       |                                       | 5.00 pm                                                 | -                              |
| <b>Minutes</b>      |                                       |                                                         |                                |
| <b>No.</b>          | <b>Item Discussed</b>                 | <b>Details</b>                                          | <b>Person-In-Charge</b>        |
| 1.                  | Task 5 discussion                     | All members were assigned to read task 5 thoroughly     | All members                    |
| 2.                  | Task 5 discussion                     | Ain shared the information about task 5 and its rubric. | Anis Safiyya and Ain Nurnabila |
| 3.                  | Problem solving                       | Solve the problems through discussion                   | Damiya Aina                    |
| 3.                  | Suggestion for subnetworking division | Each member suggest their preferred subnetwork divison  | All members                    |

## Meeting Minutes #2

|               |                                                             |                                                                                                  |                    |
|---------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------|
| Date/Time     |                                                             | 26 Dis 2024 / 9.30 pm                                                                            |                    |
| Location      |                                                             | Foyer Dewan Resak                                                                                |                    |
| Agenda        |                                                             | Progress Discussion                                                                              |                    |
| Meeting Host  |                                                             | Damiya Aina binti Basir Abd Shammad                                                              |                    |
| Attendance    |                                                             |                                                                                                  |                    |
| Name          |                                                             | Time                                                                                             | Reason For Absence |
| Anis Safiyya  |                                                             | 9.30 pm                                                                                          | -                  |
| Nurul Asyikin |                                                             | 9.30 pm                                                                                          | -                  |
| Damiya Aina   |                                                             | 9.30 pm                                                                                          | -                  |
| Ain Nurnabila |                                                             | 9.30 pm                                                                                          | -                  |
| Minutes       |                                                             |                                                                                                  |                    |
| No.           | Item Discussed                                              | Details                                                                                          | Person-In-Charge   |
| 1.            | Sharing session on current progress                         | All of the members shared the current progress                                                   | All members        |
| 2.            | Discuss the divided subnetwork and its IP addressing scheme | - All members share ideas and opinions of IP addressing scheme that will be used for the project | All members        |
| 3.            | Final edit in report                                        | - Each member completing the additional information in report                                    | All members        |
| 4.            | Meeting ended                                               | At 11.00                                                                                         | Damiya Aina        |